Evidence-Based Research Assistance

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This document is an illustrative input, not a published research study.

A research assistant can answer questions using evidence from uploaded PDFs.

The workflow has three stages: indexing, retrieval, and generation.

1. Indexing: Extract the PDF text, split it into chunks, and create embeddings.

Store each embedding with its source text and paper title in a vector database.

2. Retrieval: Embed the question and search for passages with similar meaning.

Filter by the selected domain so the answer uses relevant research passages.

3. Generation: Supply the retrieved passages to a language model.

The model should answer using those passages and acknowledge missing evidence.

Paper titles shown with the answer help the reader trace the source evidence.

References are excluded from indexing to reduce irrelevant search results.

References

Bibliography-only sentinel: this line must not be indexed.